3.2 Unit 2 INFO2 Living in the Digital World

Today's students are living in a world where the use of ICT surrounds them, and where they, and others, frequently take this use for granted.

It is increasingly important for future adults to be aware of the numerous issues arising from the use of ICT for individuals, society and organisations. The issues change rapidly and increasingly involve environmental and ethical ones.

This unit is designed to give students the wider picture of the use of ICT and to enable the understanding of basic terms and concepts involved in the study of the subject. Students should be able to discuss and comment on issues from a position of knowledge and they can do this only if they have the knowledge and understanding that underpins the subject.

Students should be encouraged to consider the important issues involved in the use of ICT by themselves and by others. Students should also study the immediate effect on themselves and also the longer term effects on society and the world in general.

Students should be asking questions such as:

- why is ICT being used
- is it appropriate to use ICT
- what are the implications of its use for me, now and in the future
- how does a particular use of ICT affect society?

It is expected that students will study much of the content of this unit through the investigation of examples of ICT use, and the issues involved in those examples.

The Teacher Resource Bank will provide further amplification and a glossary of terms.

Topic	**Key Concepts**	**Content and Amplification**
1. An ICT system and its components	What is ICT?	The use of technology for the input, storage, processing and transfer of data and the output of information.
	What is a system?	Any system involves input, processing and output.
	What is an ICT system?	ICT systems are those where the output from the system goes directly to a human being or into another ICT system.
	Any ICT system is made up of components.	The components are: • people • data • procedures • software • hardware • information.

Topic	Key Concepts	Content and Amplification
2. Data and information	Data	Define data. How data can arise. Data can be in different forms including text, still and moving images, numbers and sound. The need to code data on collection to enable effective processing. The need to encode data on input to an ICT system.
	Data is processed to produce information.	Processing is required to enable information to be produced from data.
	Information	Define information. Factors that affect the quality of information.
3. People and ICT systems	ICT systems are designed for and used by people. ICT systems are commissioned for a purpose.	
	Characteristics of users.	Different users have differing requirements dependent upon: • experience • physical characteristics • environment of use • task to be undertaken • age.
	How users interact with ICT systems.	Students should be able to discuss the need for effective dialogue between humans and machines. Appropriate interface design to provide effective communication for users. The need for the provision of appropriate help and support for users of ICT systems. The benefits and limitations of different types of user interfaces.
	Working in ICT.	Students should be aware of the many different jobs available to ICT professionals. The personal qualities and general characteristics necessary for an ICT professional to work effectively within the industry. The characteristics of an effective ICT team.

Topic	Key Concepts	Content and Amplification
4. Transfer of data in ICT systems	Students should be aware of current and emerging communication technologies.	
	Basic elements of an ICT network.	Students should be aware of what is available in order to create and use an ICT network: • communication devices • networking software • data transfer media • standards and procedures • ICT networks for different geographical scales and uses should be considered. See the Teacher Resource Bank.
	Characteristics of networks.	The World Wide Web and the Internet and the ability to distinguish between them. The characteristics of intranets and extranets.
	Uses of communication technologies.	Candidates should be able to comment on the appropriate and inappropriate use of networks and other technologies for a range of activities.
	Standards	The need for standards when transferring data.
5. Safety and security of data in ICT systems	The need to protect data in ICT systems.	Students should be able to discuss issues involving the: • privacy of data in ICT systems • legislation to protect data • commercial and intrinsic value of data.
	Threats to ICT systems.	There are different threats to ICT systems, including: • internal and external threats • malpractice and crime.
	Protecting ICT systems.	The measures that can be taken to try to protect all parts of ICT systems, including networks, against threats: • hardware measures • software measures • procedures.
	Legislation	Students should be aware of the provisions for legal action to be taken in the event of security breaches. The current legislation.

Topic	Key Concepts	Content and Amplification
6. Backup and recovery	The need for regular and systematic backup.	Case studies can be used to illustrate effective strategies and the problems of not having backup for individual users and groups of users
	Different individuals and organisations have different needs for backup and recovery.	These depend on the uses which are made of ICT. Options for backup and recovery dependent upon: • what • when • how • storage, including media and location.
	Responsibility needs to be allocated for backup and recovery procedures. The need for continuity of service.	Examples where continuity of service is different, including where different types of processing are used.
7. What ICT can provide	What ICT can provide.	ICT can provide: • fast, repetitive processing • vast storage capability • the facility to search and combine data in many different ways that would otherwise be impossible • improved presentation of information • improved accessibility to information and services • improved security of data and processes.
	Is the use of ICT systems always appropriate?	Students should be able to discuss ICT systems that: • have limitations in what they can be used for • have limitations in the information that they produce • do not always provide the most appropriate solutions.
	Types of processing.	There are different types of processing: • batch • interactive • transaction. (See Teacher Resource Bank for definitions.) Characteristics of each type. Appropriate contexts for their use.
8. Factors affecting the use of ICT	How the use of ICT is influenced by the following factors: • cultural • economic • environmental • ethical • legal • social.	Students should be able to discuss current issues, using examples to illustrate where and how each of the factors has had an effect on the use of ICT. See Teacher Resource Bank for ideas and resources.
9. The consequences of the use of ICT	For individuals. For society.	Students should be able to discuss, using examples, the consequences of the use of ICT for different groups of individuals and society as a whole. See Teacher Resource Bank for ideas and resources.